

Configure API-KEY

Configure API-KEY

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1. Course Content

- Configure the robot's in-vehicle API-KEY using the previously registered API-KEY

[!WARNING]

Note: Please ensure the in-vehicle system is connected to the internet to use the cloud-based model services.

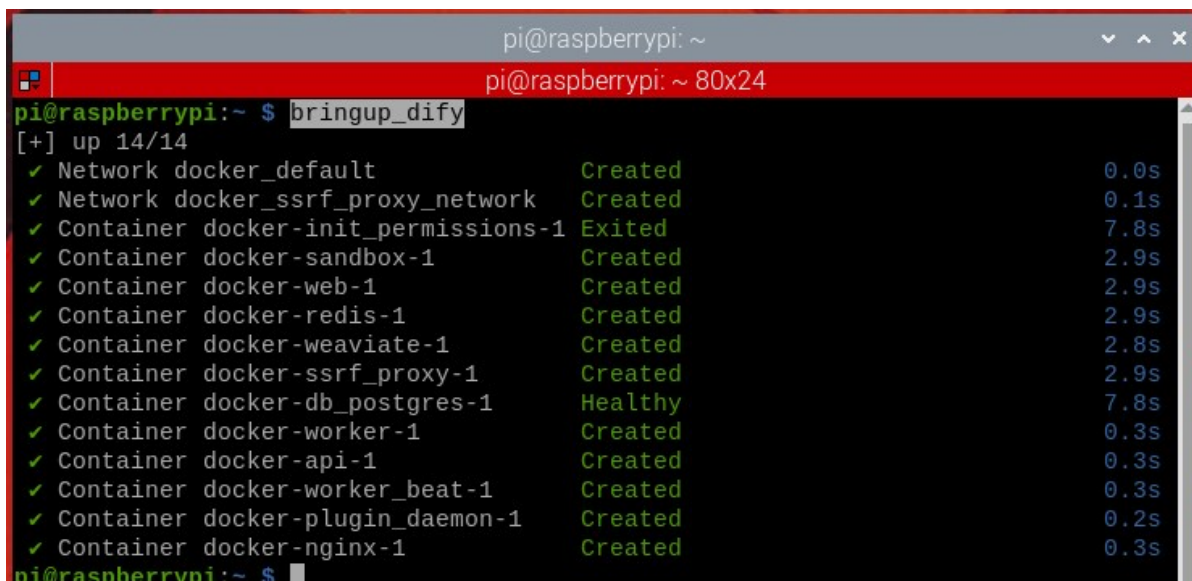
2. Start the Dify Service

[!TIP]

MutoRS uses Dify to build a multi-agent system, with Dify centrally managing the invocation of cloud-based models.

- Connect to the vehicle's infotainment system via VNC or SSH. Enter the following command in the terminal:

```
bringup_dify
```



```
pi@raspberrypi: ~  
pi@raspberrypi: ~ 80x24  
pi@raspberrypi:~ $ bringup_dify  
[+] up 14/14  
✓ Network docker_default Created 0.0s  
✓ Network docker_ssrf_proxy_network Created 0.1s  
✓ Container docker-init_permissions-1 Exited 7.8s  
✓ Container docker-sandbox-1 Created 2.9s  
✓ Container docker-web-1 Created 2.9s  
✓ Container docker-redis-1 Created 2.9s  
✓ Container docker-weaviate-1 Created 2.8s  
✓ Container docker-ssrf_proxy-1 Created 2.9s  
✓ Container docker-db_postgres-1 Healthy 7.8s  
✓ Container docker-worker-1 Created 0.3s  
✓ Container docker-api-1 Created 0.3s  
✓ Container docker-worker_beat-1 Created 0.3s  
✓ Container docker-plugin_daemon-1 Created 0.2s  
✓ Container docker-nginx-1 Created 0.3s  
pi@raspberrypi:~ $
```

- View the vehicle's infotainment system's IP address. This can be done via the OLED screen or in the terminal using `ifconfig`.

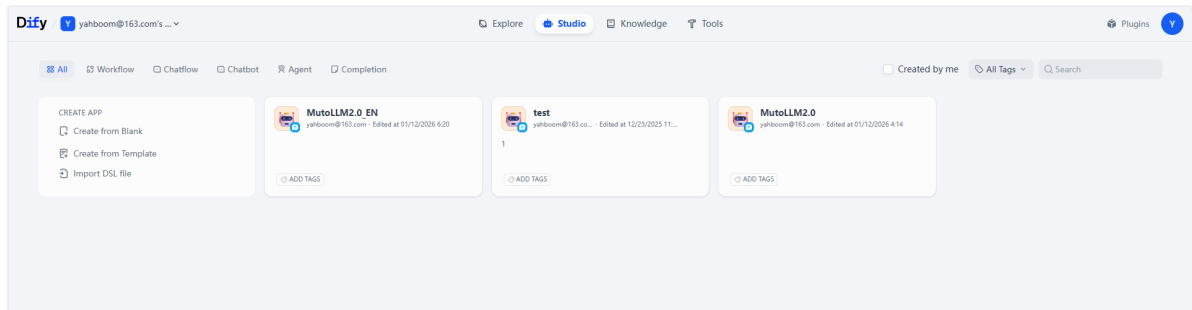
```
wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.12.51 netmask 255.255.255.0 broadcast 192.168.12.255
    inet6 fe80::a983:ca12:d35d:6436 prefixlen 64 scopeid 0x20<link>
    ether 2c:cf:67:ec:a6:20 txqueuelen 1000 (Ethernet)
    RX packets 20830 bytes 6225753 (5.9 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 11913 bytes 11213665 (10.6 MiB)
    TX errors 0 dropped 2 overruns 0 carrier 0 collisions 0

pi@raspberrypi:~$
```

- Directly enter the vehicle's infotainment system's IP address in your browser's address bar to access the Dify management page. If this is the first time logging in, a username and password are required. You can select the language in the upper left corner.

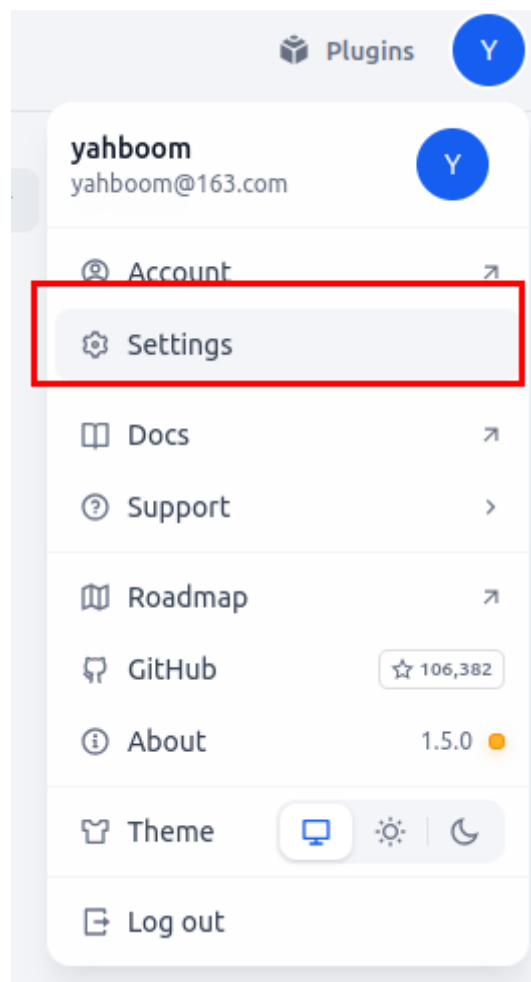
[!NOTE]

- Username: `yahboom@163.com`
- Password: `yahboom123`
- All account passwords, agent applications, and RAG data are stored locally.
- After logging in, see the image below.

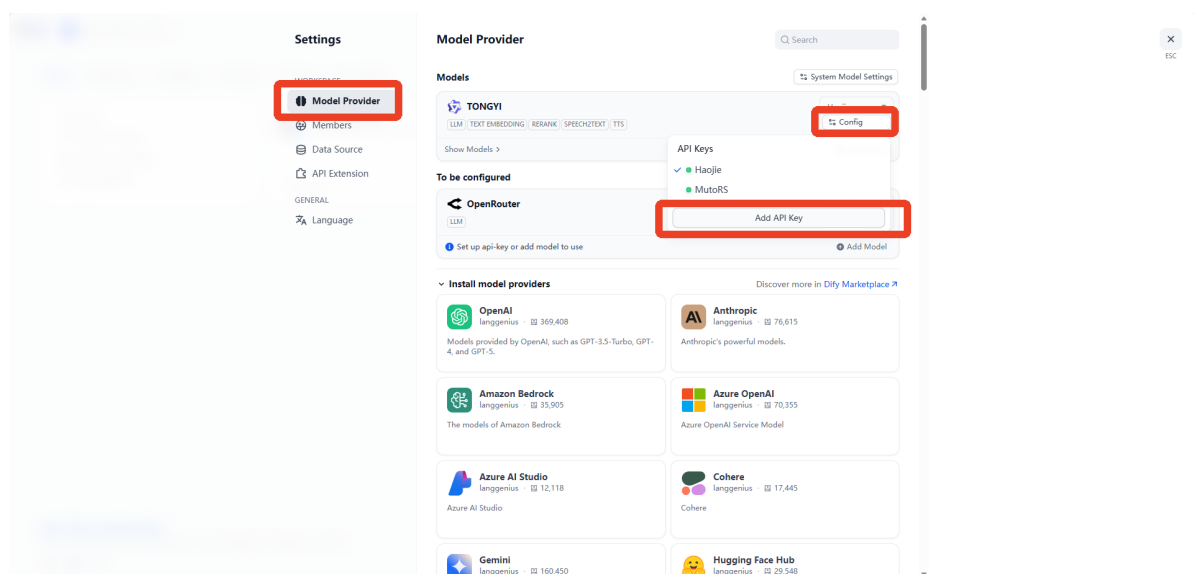


3. Configure Model Service Provider API-KEY

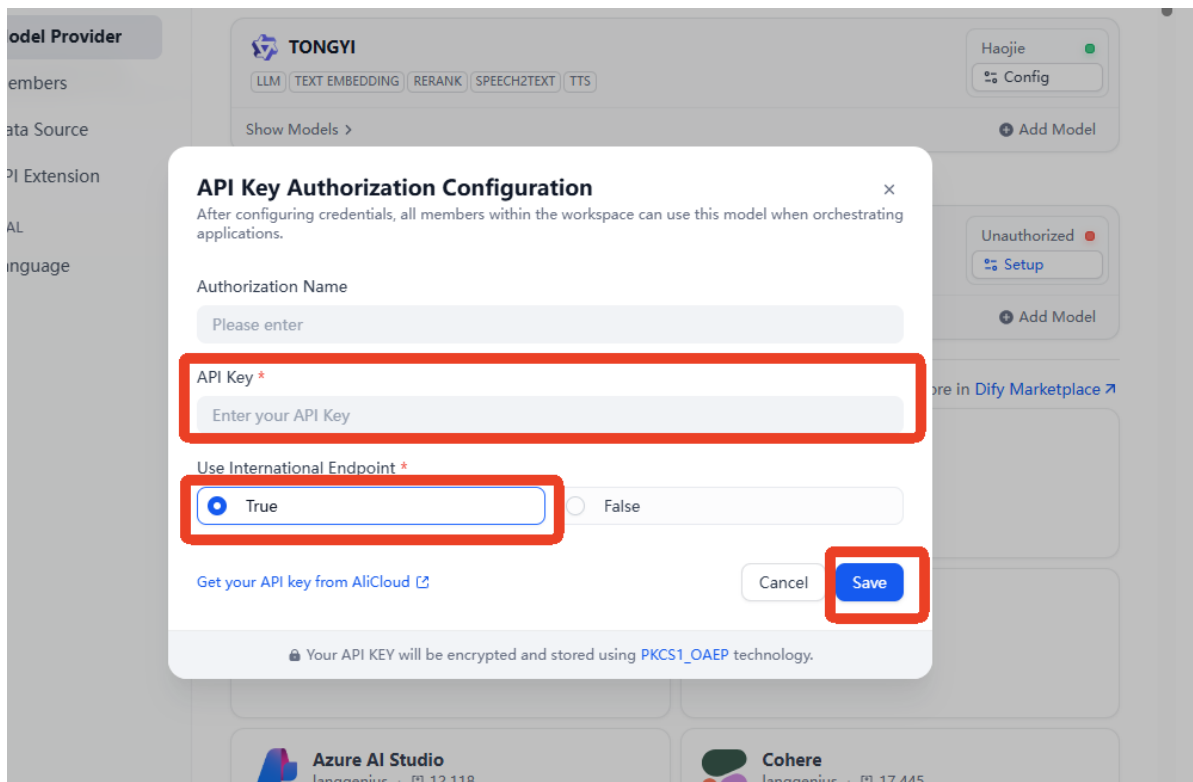
- Click `Settings`



- Here, we take configuring the API for a Balian account as an example. Click **Model Provider** -> **Config**-> **Add API Key**



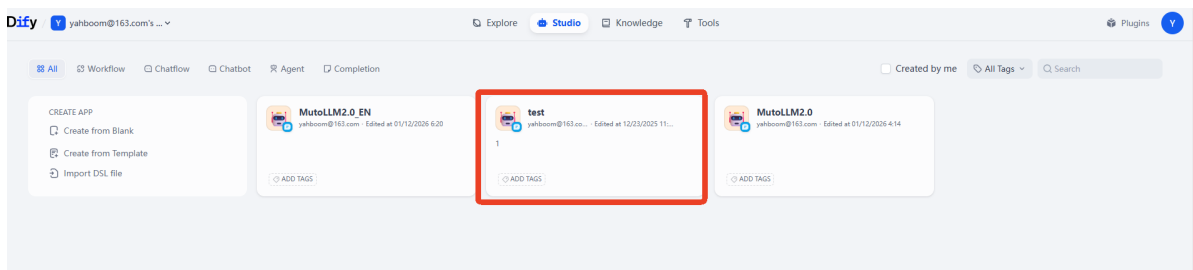
- Enter your Balian platform's API-KEY, then select whether it is an international battle account, and click Save.



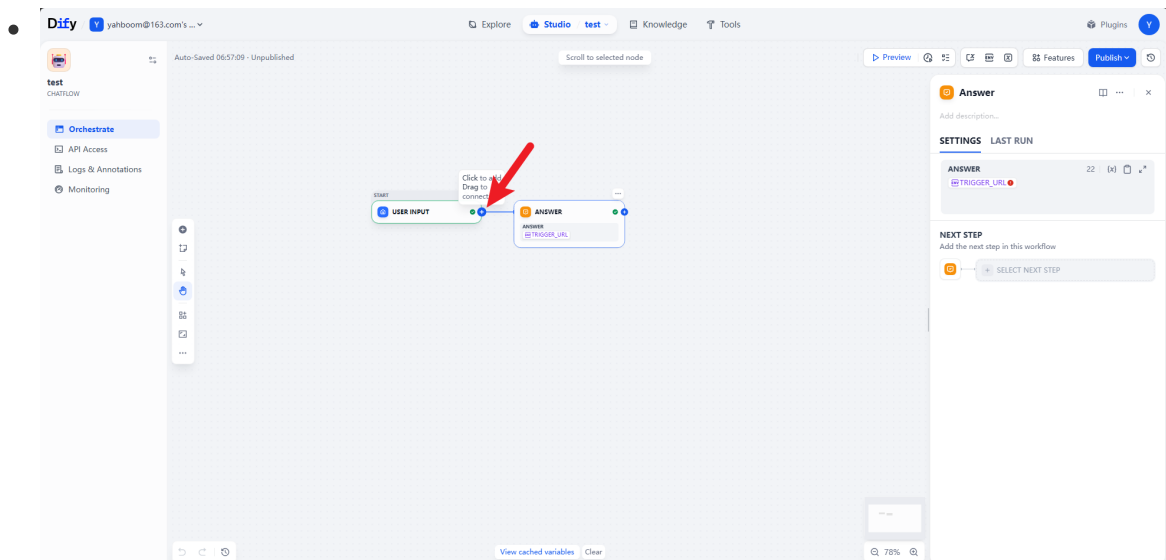
4. Testing the API Key

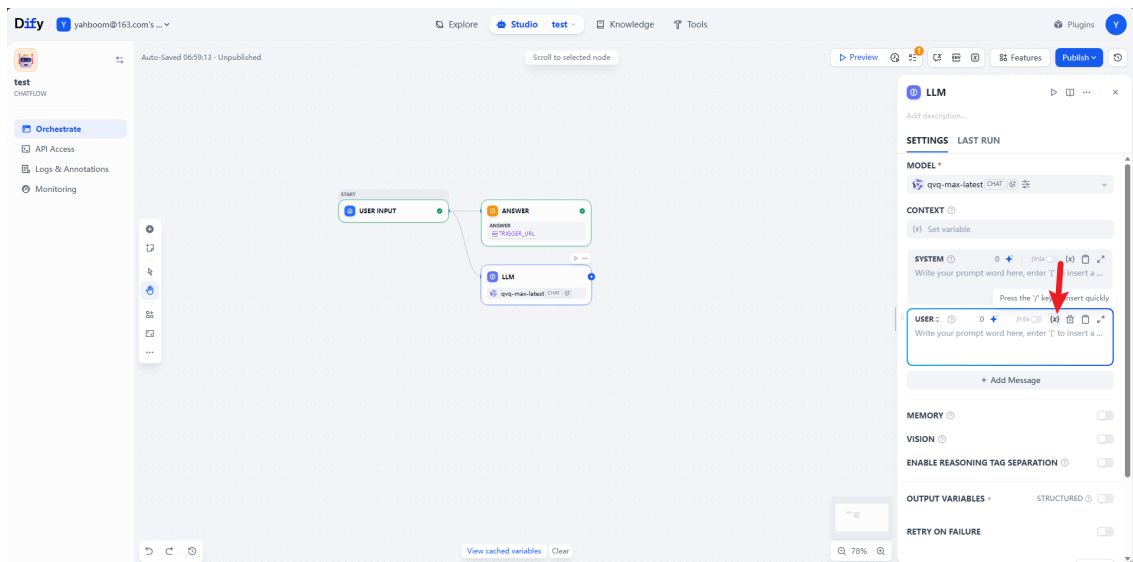
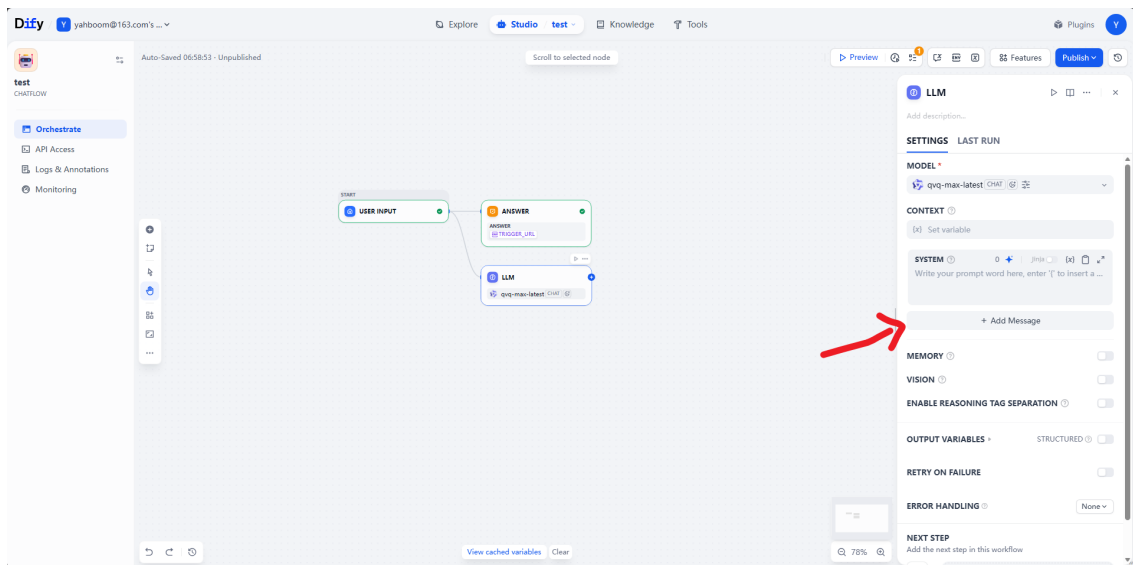
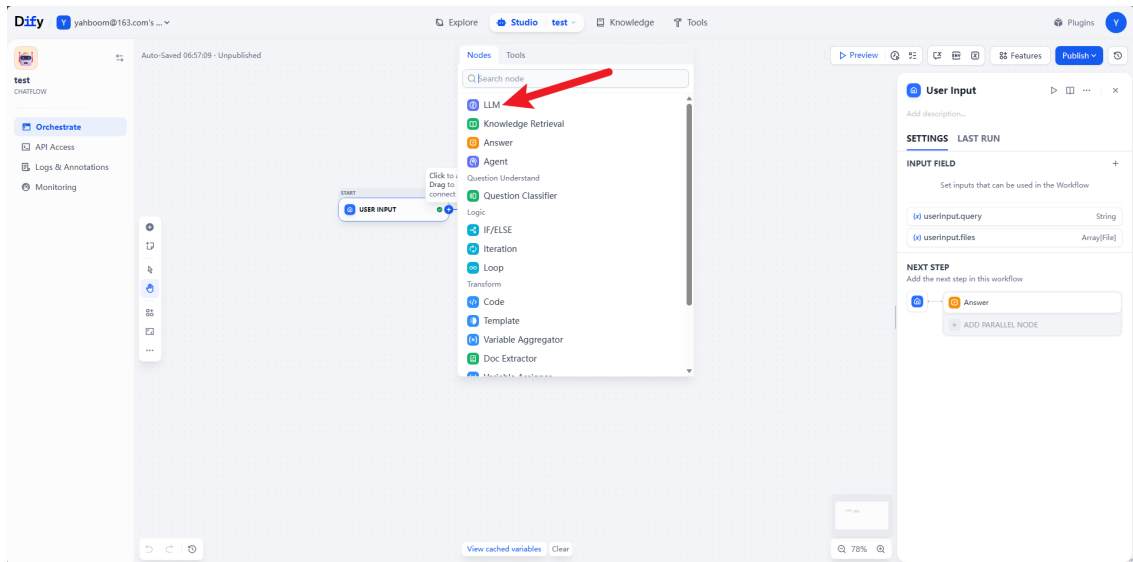
[!TIP]

- If you need to test whether the API key is available, you can refer to this section of the tutorial. If not, you can ignore this step.
- Click the TEST application in the studio.



- Then, select any model from the model selection menu to test.





Diffy yahboom@163.com's ...

Explore Studio test Knowledge Tools

Auto-Saved 06:59:34 - Unpublished

Scroll to selected node

Preview Features Publish

test CHATFLOW

Orchestrate

API Access

Logs & Annotations

Monitoring

START

USER INPUT

ANSWER

LLM

qyq-max-latest CHAT

LLM

qyq-max-latest CHAT

SETTINGS LAST RUN

MODEL

qyq-max-latest CHAT

CONTEXT

Set variable

SYSTEM

Write your prompt word here, enter 'T' to insert a ...

Press the 'I' key to insert quickly

USER

1

Search variable

id query String

files Array/File

SYSTEM

sysdialogue_count Number

sysconversation_id String

sysuser_id String

sysapp_id String

sysworkflow_id String

sysworkflow_run_id String

Context

View cached variables Clear

78%

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Scroll to selected node

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Orchestrate

API Access

Logs & Annotations

Monitoring

START

USER INPUT

ANSWER

LLM

qyq-max-latest CHAT

LLM

qyq-max-latest CHAT

SETTINGS LAST RUN

MODEL

qyq-max-latest CHAT

CONTEXT

Set variable

SYSTEM

Write your prompt word here, enter 'T' to insert a ...

USER

15

User Input (id query)

Add Message

MEMORY

VISION

ENABLE REASONING TAG SEPARATION

OUTPUT VARIABLES

STRUCTURED

RETRY ON FAILURE

View cached variables Clear

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test CHATFLOW

Orchestrate

API Access

Logs & Annotations

Monitoring

START

USER INPUT

ANSWER

LLM

qyq-max-latest CHAT

ANSWER

ANSWER

TRIGGER_URL

delete this

NEXT STEP

Add the next step in this workflow

SELECT NEXT STEP

View cached variables Clear

78%

Dify | yahboom@163.com's ...

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Preview

Plugins

test CHATFLOW

Orchestrate

API Access

Logs & Annotations

Monitoring

START

USER INPUT

LLM

ANSWER

View cached variables

Clear

78%

Answer

Add description...

SETTINGS LAST RUN

Press the / key to insert quickly

ANSWER

Write your prompt word here, enter / to insert a var...

NEXT STEP

Add the next step in this workflow

SELECT NEXT STEP

Dify | yahboom@163.com's ...

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Scroll to selected node

Preview

Plugins

test CHATFLOW

Orchestrate

API Access

Logs & Annotations

Monitoring

START

USER INPUT

LLM

ANSWER

View cached variables

Clear

78%

Answer

Add description...

SETTINGS LAST RUN

ANSWER

Search variable

LLM

text String

reasoning text String

usage Object

USER INPUT

query String

files Array[File]

SYSTEM

sys.dialogue_count Number

sys.conversation_id String

sys.user_id String

sys.app_id String

Dify | yahboom@163.com's ...

Explore Studio test Knowledge Tools

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Scroll to selected node

Preview

Plugins

test CHATFLOW

Orchestrate

API Access

Logs & Annotations

Monitoring

START

USER INPUT

LLM

ANSWER

View cached variables

Clear

78%

Answer

Add description...

SETTINGS LAST RUN

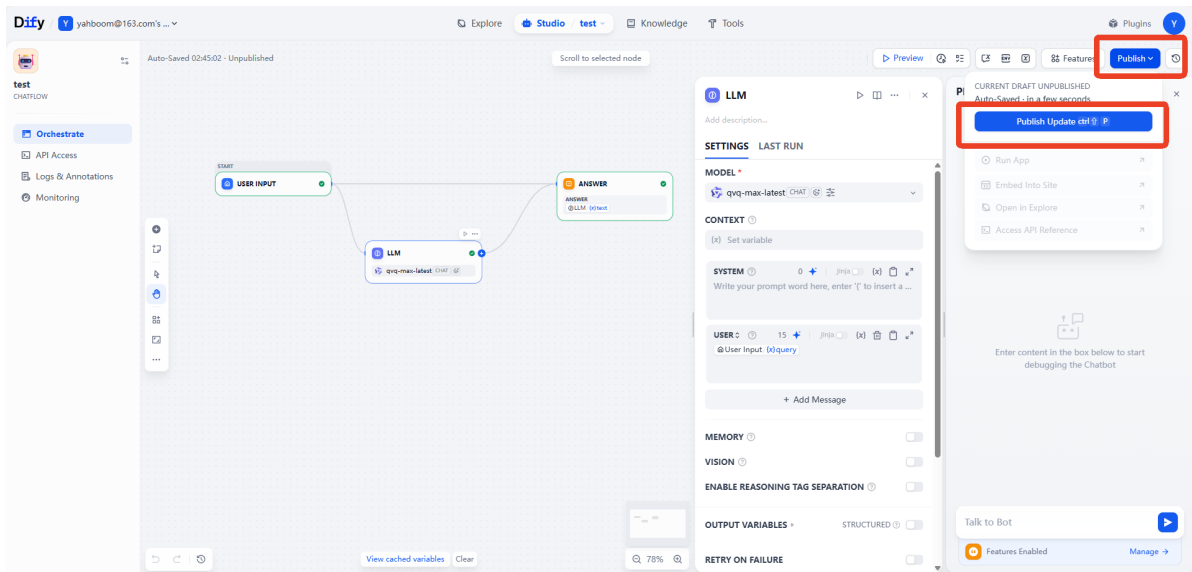
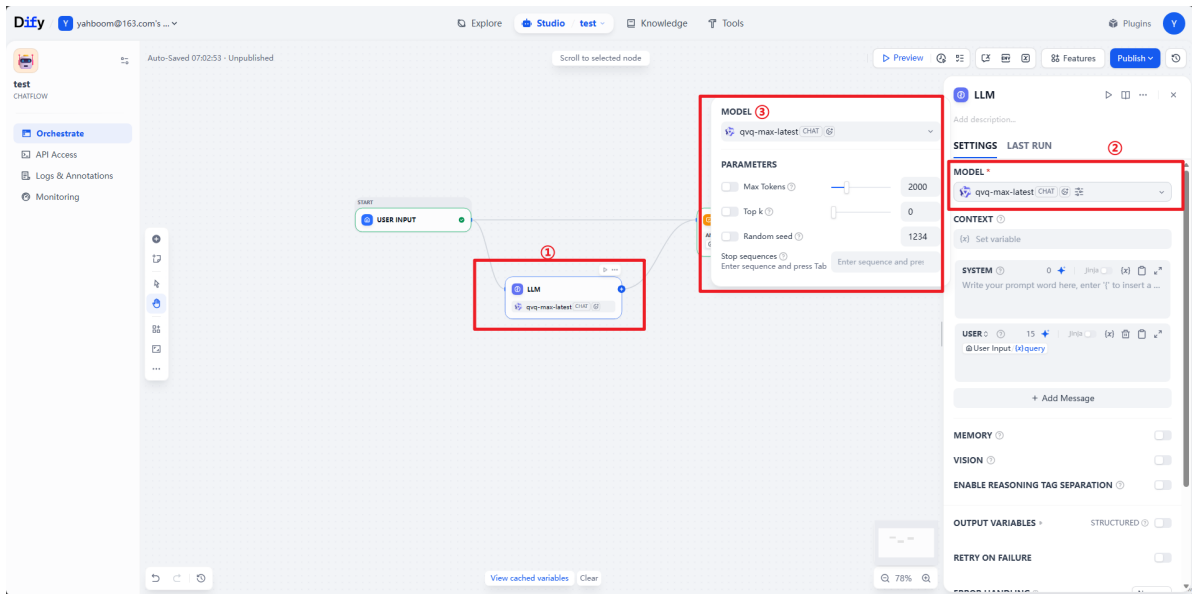
ANSWER

@LLM [text]

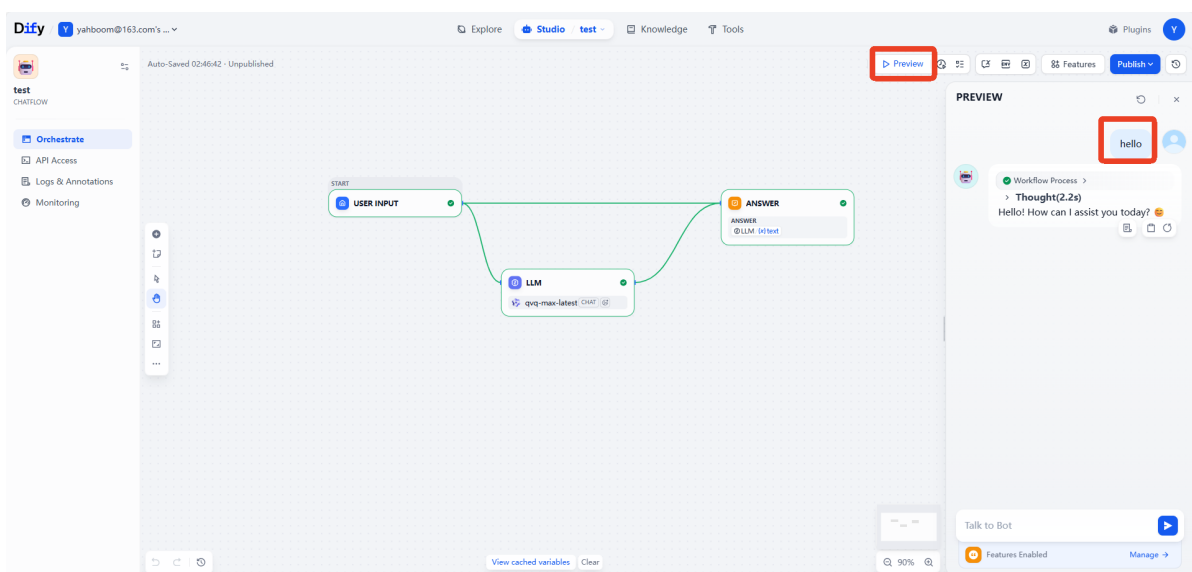
NEXT STEP

Add the next step in this workflow

SELECT NEXT STEP



- Enter any content in the chat box. If the registered API-KEY is valid, you will be able to see the model's reply content normally.



5. Configuring the muto-llm (MutoRS) API Package

The configuration of MutoRS is slightly different from other products.

Here we need to prepare the API Key for the Alibaba Cloud Model Studio Platform, in the format "sk-xxxxxxxxxxxxx".

In VNC, enter the Raspberry Pi motherboard.

Enter the following command:

```
sudo gedit /usr/bin/start_muto_docker
```

```
1#!/usr/bin/env bash
2
3# =====
4# User Configuration Area
5# =====
6
7# 1. Voice Module Configuration
8# =====
9
10# 1.1 Provider & Keys
11# -----
12export ALIBABA_API_KEY="sk-XXXXXXXXXXXX20"
13
14export FORCE_VOICE_PROVIDER="xunfei"
15
16# 1.2 General Voice Settings
17# -----
18export VOICE_LANGUAGE="en"
19
20# 2. Dify Configuration
21# Auto-detect IP from wlan0
22export DIFY_HOST=$(ip addr show wlan0 2>/dev/null | grep "inet " | awk '{print $2}' | cut -d/ -f1 | head -n 1)
23# Fallback if wlan0 has no IP
24if [ -z "$DIFY_HOST" ]; then
25    export DIFY_HOST="127.0.0.1"
26fi
```

①: Enter your API here. Key

②: Language selection, set to zh_cn for Chinese, en for English

Overseas users need to change this configuration item to "xunfei"

```
3# =====
4# User Configuration Area
5# =====
6
7# 1. Voice Module Configuration
8# =====
9
10# 1.1 Provider & Keys
11# -----
12export ALIBABA_API_KEY="sk-XXXXXXXXXXXX"
13
14export FORCE_VOICE_PROVIDER="xunfei"
15
16# 1.2 General Voice Settings
17# -----
18export VOICE_LANGUAGE="zh_cn"
19
20# 2. Dify Configuration
21# Auto-detect IP from wlan0
22export DIFY_HOST=$(ip addr show wlan0 2>/dev/null | grep "inet " | awk '{print $2}' | cut -d/ -f1 | head -n 1)
23# Fallback if wlan0 has no IP
24if [ -z "$DIFY_HOST" ]; then
25    export DIFY_HOST="127.0.0.1"
26fi
```

After setting, click the save button in the upper right corner to save the script.